Opinionated Harnesses

Building AI Agents That Actually Work

Riccardo Pirruccio

github.com/RPirruccio/claude-mem

Austin LangChain AI Meetup @ Cloudflare

About Me



Ricky Pirruccio

Technical Project Manager @ Applied Materials

- Day job: project manager at a semiconductor company
- Building AI tools to track parts, manage risk, automate reporting
- At work: GitHub Copilot + VS Code with skills, MCP, CLI tooling
- Personal projects: Claude Code plugins like this one

Part of AIMUG (Austin's AI community) for 2.5 years

github.com/RPirruccio

The Engineering Evolution

v1 Prompt Eng.

craft the perfect prompt

v2 Context Eng.

what does the agent know

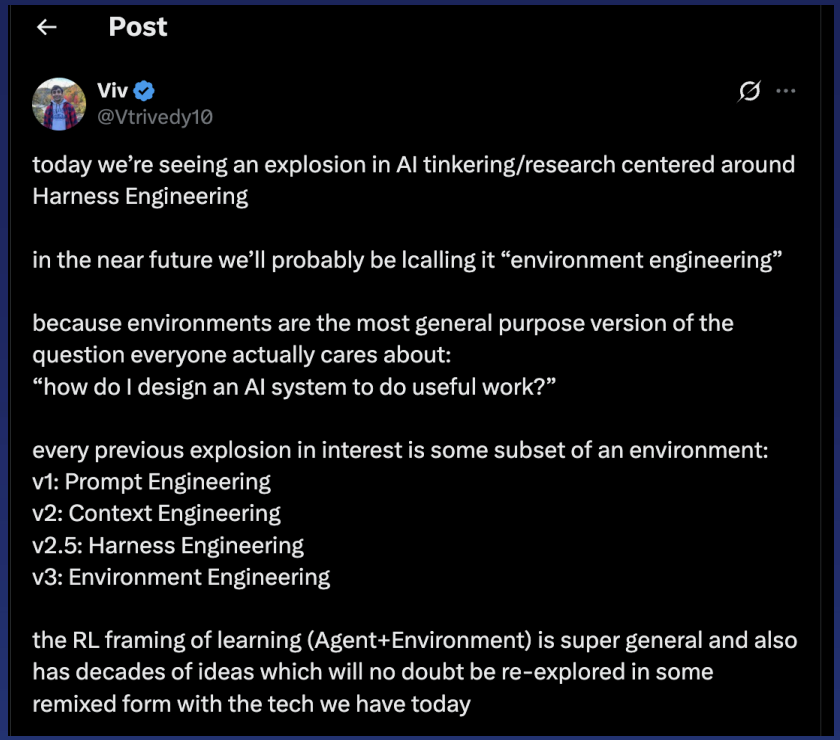
v2.5 **Harness Eng.** WE ARE HERE

hooks, tools, opinions

v3 Environment Eng.

safe sandbox, let it learn

— @Vtrivedy10



The Formula

Agent = Opinionated Harness + Model

The Harness:

- Prompts
- Tools
- Lifecycle hooks
- Context management

"Context engineering is effectively the #1 job of engineers building AI agents."

— Cognition (Devin team), June 2025

"Do we need this setting?"

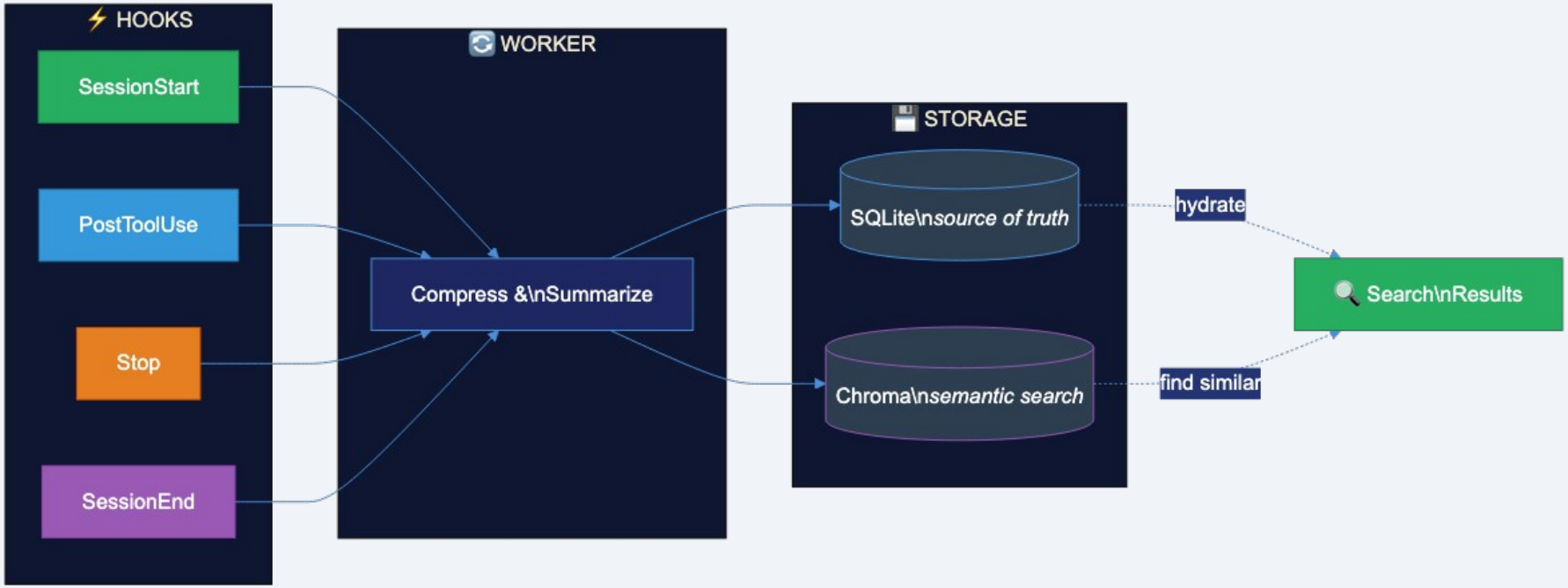
"Can we kill it?"

— Cursor's design philosophy

Every config option you expose is a decision you failed to make.

claude-mem: An Opinionated Memory Plugin

Captures what you do → Compresses with AI → Injects into future sessions



"What did I work on last week?" actually works. — Cognition's solution: compress history into key details.

The Harness: Hooks + MCP

Lifecycle events + tool exposure

Hook	When it fires	claude-mem uses it for
SessionStart	Session begins	Inject relevant past context
UserPromptSubmit	You send a message	Capture user prompts
PostToolUse	After any tool runs	Record observations (the big one)
Stop	Claude finishes response	Trigger compression/summary
SessionEnd	Session closes	Final cleanup

+ MCP (Model Context Protocol) exposes tools like search, timeline, get_observations

Hooks control the lifecycle. MCP exposes tools. Together = the harness.

How the Memory Actually Works

SQLite (source of truth)

- Stores ALL data
- Full text, metadata
- Structured queries
- Fast filters

Chroma (semantic index)

- Stores vector embeddings
- "Meaning" as numbers
- Similarity search
- "Find related things"

Search Flow:

1. Chroma finds semantically similar → returns IDs 2. SQLite hydrates full records → returns data

Like a librarian (Chroma) who knows where related books are, and a filing cabinet (SQLite) that holds them.

My Opinionated Decisions

- ✓ 5 specific lifecycle hooks (not generic callbacks)
- ✓ SQLite + Chroma for storage (not "bring your own database")
- ✓ Built-in compression (not configurable chunking)
- ✓ Local Ollama on RTX 4090 (not cloud APIs)

Your observations shouldn't leave your machine. Every file, every command — that's sensitive.

What Happens When You're NOT Opinionated?

MOLTBOOK

Feb 2, 2026

90k GitHub stars in weeks

"Social network for AI agents"

72 hours later:

- 1.5M API keys exposed
- 6,000+ emails leaked
- Anyone could hijack any agent

Root cause: Disabled row-level security on Supabase

The Bigger Picture

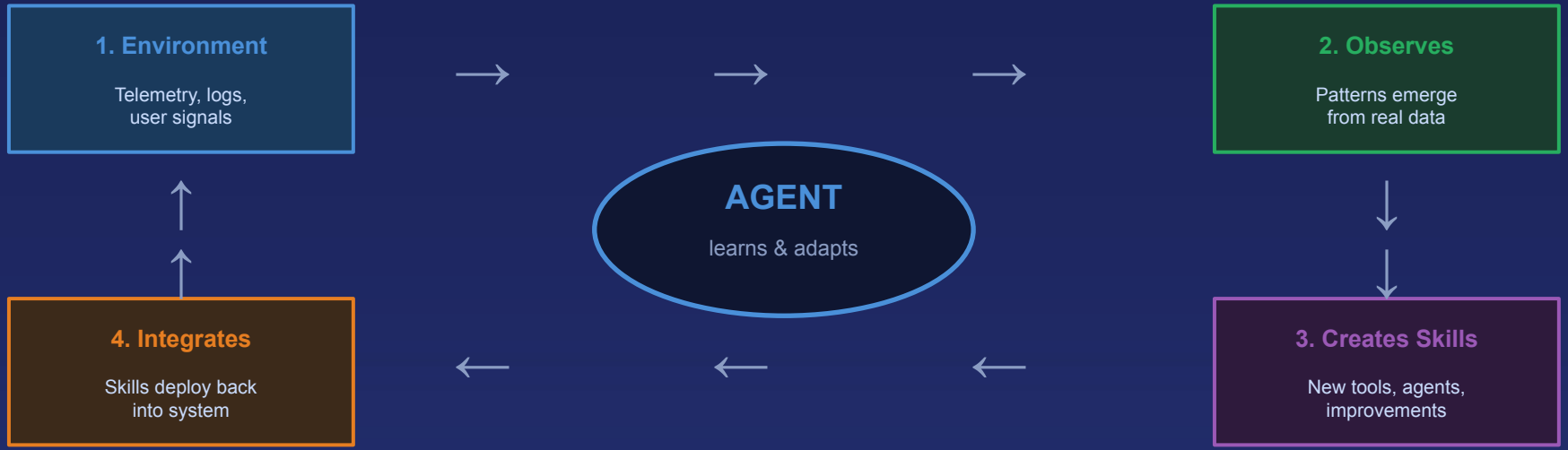
AI tools are shipping faster than security can keep up

Tool	Issue	Date
Moltbook	**1.5M API keys exposed**	Feb 2, 2026
ClawHub	**341 malicious skills**	Feb 4, 2026
Cursor	**84% prompt injection success**	Research
5 platforms	**69 vulnerabilities**	Jan 2026

Vibe coding is shipping faster than security can keep up. The harness is your defense until v3 arrives.

But There's Hope: V3 is Coming

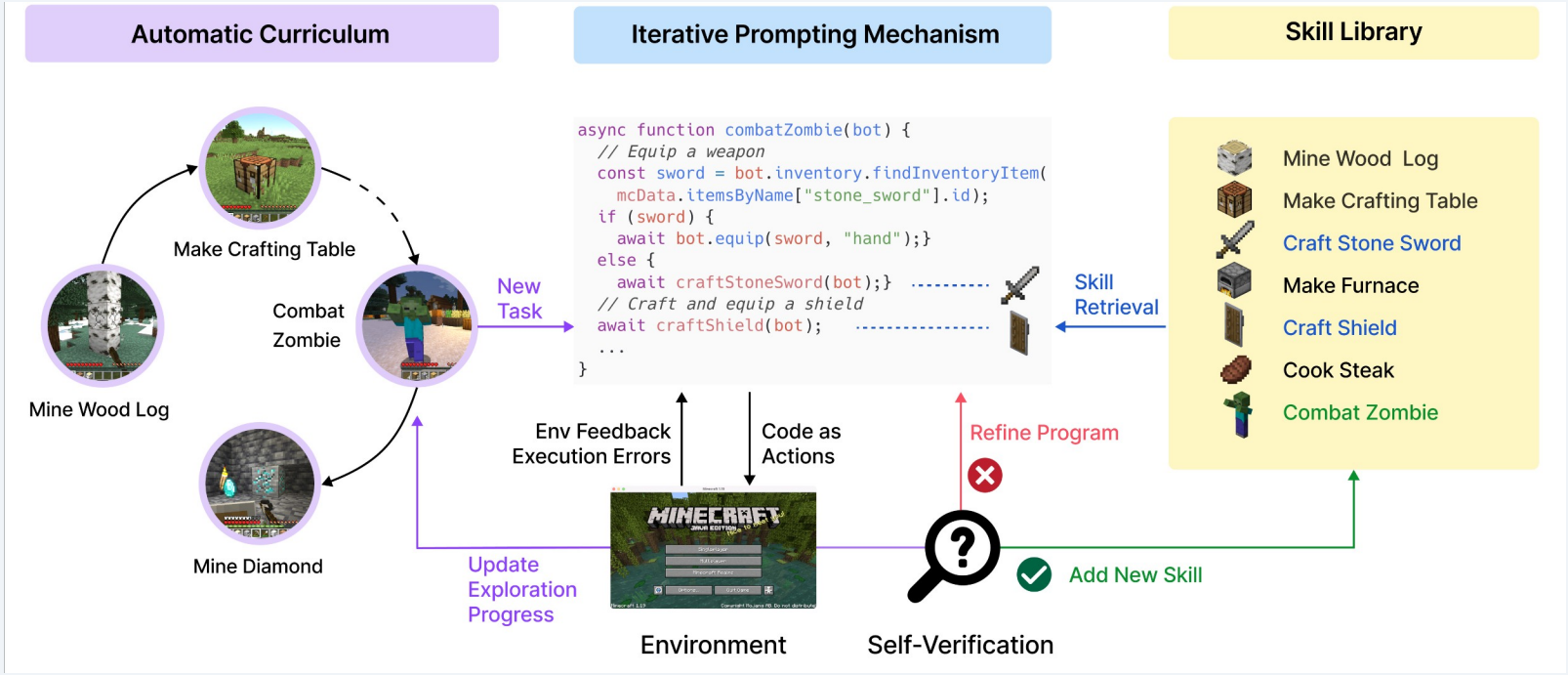
The Environment Loop



Toyota + Voyager are doing this NOW. Infinite compounding. This is where we're headed.

Where This Pattern Comes From

Voyager (2023): GPT-4 learned Minecraft by building a reusable skill library



The Voyager Loop

Automatic curriculum in action



The prompt was basically:

```
"You are in Minecraft.
Here's your inventory: [wooden pickaxe]
Here's what you see: [trees, cave]
Skills you have: [chopTree(), craftPickaxe()]
What should you explore next?"
```

GPT-4 already knows Minecraft's tech tree.

It proposes its own goals. No human in the loop for task selection.

Build Opinionated Harnesses (for now)

1. **Narrow and deep > wide and shallow**
   Pick a task. Make it work reliably.

2. **Make decisions so users don't have to**
   Every config option is a failure.

3. **Keep your context local**
   Your observations are sensitive data.

4. **Know the limits**
   V3 is coming. Until then, the harness is your defense.

Questions?

github.com/RPirruccio/claude-mem

Viv's blog: vtrivedy.com

Cognition: cognition.ai/blog/dont-build-multi-agents

Voyager: voyager.minedojo.org